

Weeks 1 and 2 Assignment DUE

Assignment Goal/Objective: To develop a spreadsheet-based decision model that balances the costs of overcapacity (waste) against the costs of undercapacity (lost revenue/service).

Assignment (For Business students) – Forecasting Coffee Shop Muffin Sales. [Scroll down for Healthcare Administration Assignment]

Objective: Develop a spreadsheet-based decision model to find the "Optimal Order Quantity" by balancing the cost of lost sales against the cost of wasted inventory.

Scenario:

You are the manager of Morning Brew Café, a coffee shop in a busy downtown area. One of your most popular items is your fresh blueberry muffins, baked daily. You must decide how many muffins to bake each morning.

The fixed daily baking setup cost (regardless of the number baked) is \$40. The variable cost per muffin (ingredients, labor, packaging) is \$1.20. The selling price per muffin is \$3.50 during the morning and lunchtime rush (6:30 a.m. – 2:00 p.m.).

After 2:00 p.m., any remaining muffins are sold at a discounted price of \$1.00 each to clear inventory before closing.

Based on past data, your best guess for demand during full-price hours is 180 muffins per day. However, actual demand varies daily because of factors like weather and foot traffic.

You are considering baking 170 muffins per day, but you want to build a spreadsheet model to:

- Experiment with uncertain demand.
- Change the order quantity and see the profit impact.
- Understand the trade-offs between understocking (missed sales) and overstocking (leftover muffins).

The Financial Logic

- **Fixed Daily Setup Cost:** \$40.00
- **Variable Cost (Unit Cost):** \$1.20
- **Selling Price (Full):** \$3.50
- **Salvage Value (Discount Price):** \$1.00

Assignment Questions:

1. Model Setup

Create a spreadsheet model that calculates **Daily Profit**. Your formulas must account for the fact that you cannot sell more than you bake, and you cannot sell more than the customers demand.

- **Full Price Sales** = $\text{MIN}(\text{Demand}, \text{Baked})$
- **Discounted Sales (Leftovers)** = $\text{MAX}(0, \text{Baked} - \text{Demand})$
- **Total Revenue** = $(\text{Full Price Sales} * 3.50) + (\text{Discounted Sales} * 1.00)$
- **Total Cost** = $40 + (\text{Baked} * 1.20)$

Consider the following in your model set up:

- Fixed daily baking cost
- Variable baking cost per muffin
- Selling prices (full price and discount price)
- Demand quantity
- Muffins baked per day

2. Scenario Analysis

Using your model, calculate the total profit for the following scenarios if you choose to **bake 180 muffins**:

- What is the profit if Demand is only **140**?
- What is the profit if Demand is **220**?
- *Observation*: Why doesn't profit increase indefinitely as demand rises?

3. Optimization & Probability

Using the provided CSV dataset, calculate the probability of demand levels. Then, use a **Data Table** to find the "Expected Profit" for baking 160, 170, 180, 190, and 200 muffins. Use Excel's Data Table or Solver to determine the optimal number of muffins to bake per day that maximizes expected profit given the demand probabilities below:

- 150 muffins (15%)
- 170 muffins (25%)
- 180 muffins (30%)
- 190 muffins (20%)
- 200 muffins (10%)



Download the Business Practice Data

This dataset contains 100 days of muffin demand, influenced by "Foot Traffic" and "Day of the Week." Students should use this to build their frequency distribution.



muffin_shop_demand.csv



Raw Data Preview

Date	Day_of_Week	Foot_Traffic_Index	Actual_Muffin_Demand
2024-09-01	Monday	High	192
2024-09-02	Tuesday	Medium	174
2024-09-03	Wednesday	Medium	168
2024-09-04	Thursday	High	205
2024-09-05	Friday	Very High	215



Step-by-Step Spreadsheet Synthesis

Step 1: Calculate Probabilities. You should create a Pivot Table from the CSV.

- **Rows:** Actual_Muffin_Demand (Grouped by 10s).
- **Values:** Count of Date (Show as % of Column Total).
- *This provides the "Demand Probability" needed for the Optimization section.*

Step 2: The "What-If" Analysis Once the model is built, students should use the **Goal Seek** tool to find the "Breakeven Demand"—the minimum number of muffins they must sell at full price to cover the \$40 fixed cost and the variable costs.

Step 3: Visualizing the Trade-off Have students create a **Line Chart** showing "Quantity Baked" on the X-axis and "Expected Profit" on the Y-axis. The peak of the curve represents the mathematically optimal decision.

Assignment Questions

4. Managerial Insight

- Define the **Cost of Understocking (\$C_u\$)**: The profit lost for every muffin you *could* have sold but didn't have and discuss how your recommended baking quantity balances the risk of stockouts and overproduction.
- Define the **Cost of Overstocking (\$C_o\$)**: The net loss for every muffin you baked but had to sell at a discount. Also, suggest at least one non-price strategy to reduce leftover muffins.
- **Strategy:** Aside from discounting, how could a manager use "Bundling" or "Pre-orders" to reduce the risk of leftover muffins?

Healthcare Administration – Hospital Operations

Identify a specific healthcare service (e.g., staffing an Urgent Care clinic, ordering flu vaccines, or scheduling MRI slots) that follows this same "perishable" logic.

- What is the "Salvage Value" in your healthcare example?
- What are the ethical implications of "understocking" in a clinical setting vs. a cafe?

In this scenario, "Demand" represents patients arriving for a vaccine, and "Supply" represents the doses ordered for that day. This dataset includes 100 days of historical data, allowing students to calculate the probabilities needed for their Optimization.

Examples of components for Healthcare Administration scenario

Healthcare Component (Flu Clinic)	
Fixed Cost	Clinic overhead, electricity, and administrative setup.
Variable Cost	Cost of one vaccine dose + syringe + clinician time.
Selling Price	Insurance reimbursement rate per vaccination.
Discount Price	Partial credit for returned unused vials (if applicable).
Uncertain Demand	Number of citizens who actually show up for their appointment.
Order Quantity	Number of doses ordered or nurses scheduled for the day.

The Scenario: Metro Community Flu Clinic

- Fixed Setup Cost: \$500.00 (Staffing, facility rental, and refrigeration).
- Variable Cost per Dose: \$12.00 (Acquisition and medical supplies).
- Reimbursement (Price): \$45.00 (Standard insurance payout).
- Salvage Value: \$3.00 (Manufacturer buy-back for unused/expired vials).
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 Download the Practice Data



healthcare_vaccine_demand.csv

Raw Data Preview (First 10 Rows)

Date	Actual_Patient_Demand	Weather_Impact	Marketing_Blast_Sent
2024-10-01	162	Sunny	No
2024-10-02	178	Rain	Yes
2024-10-03	155	Cloudy	No
2024-10-04	192	Sunny	Yes
2024-10-05	148	Rain	No
2024-10-06	171	Sunny	No
2024-10-07	184	Sunny	Yes
2024-10-08	166	Cloudy	No
2024-10-09	204	Sunny	Yes
2024-10-10	175		

1. Identifying Perishable Inventory

Question: Identify a specific healthcare service (e.g., staffing an Urgent Care clinic, ordering flu vaccines, or scheduling MRI slots) that follows this same "perishable" logic. Explain why "time" or "dosage" acts as the expiration date.

2. Defining Salvage Value

Question: What is the "Salvage Value"?

- *Hint: If a surgical slot goes unused, is the salvage value \$0, or can it be used for emergency cases? If a flu vaccine expires, is there a tax write-off or a return credit?*

3. Ethical vs. Financial Trade-offs

Question: Discuss the ethical implications of "understocking" in a clinical setting vs. a café.

- If a café runs out of muffins, a customer is disappointed. If an ER "understocks" on on-call neurologists, what is the cost beyond "lost revenue"? How does this change the **Optimal**

Service Level?

4. Overcapacity vs. Undercapacity

Question: Using your spreadsheet results, identify the "Cost of Underage" ()—the cost of missing one unit of demand—and the "Cost of Overage" ()—the cost of having one unit too many.

- **Healthcare Context:** Is it more expensive for the hospital to have an empty bed () or to have a patient waiting in the hallway ()?

Assignment Tasks

Task 1: Frequency Analysis (Building the Probability Table) Before you can solve the optimization problem, you must "Synthesize" this raw data into the probabilities mentioned in Question 3 of your assignment.

1. Create a **Pivot Table** from the CSV.
2. Drag Actual_Patient_Demand to **Rows**.
3. Right-click a row value and select **Group**. Group by **10s** (e.g., 150-160, 161-170, etc.).
4. Drag Actual_Patient_Demand to **Values** and change setting to **% of Column Total**.
5. **Synthesis:** Use these percentages as your "Demand Probabilities" for the Solver model.

Task 2: Ethical Sensitivity Analysis Using the Excel Model, calculate the "Profit" if you order 170 doses but 200 patients show up.

- **The Math:** Revenue is capped at 170 sales.
- **The Healthcare "So What?":** The model shows a financial profit, but 30 patients were turned away.
- **Question:** What is the "Social Cost" of those 30 unvaccinated individuals in a community? How does this change the "Order Quantity" if the goal is **Public Health** rather than **Net Profit**?

Managerial Insight Questions (Healthcare Focused)

1. **The "Wait Time" Variable:** In healthcare, demand that isn't met often becomes a "waitlist." How would you account for the **reputation cost** of a 4-week wait time for an MRI?
2. **Strategic Flexibility:** Suggest one "non-price" strategy for a hospital to reduce "leftover" capacity.
 - *Example: Cross-training staff so a nurse can move from a quiet department to a busy one (Floating), or using "Telehealth" to fill gaps in a physical clinic schedule.*